C PROGRAMMING FROM SCRATCH · EPISODE 03

Your First C Program

From Empty Project to Running Code

KEY POINTS

- A project keeps your source files together with the settings Visual Studio needs to build them.
- Name the file `hello.c`. The extension, not the template, decides that it is compiled as C.
- Build Solution compiles and links; Ctrl+F5 builds when needed, then runs.
- The program lands in `x64\Debug`, named after the project: `first_program.exe`.

01 Create the Project

▶ VIDEO 00:41

1. **Create an empty project. Create a new project** → **Empty Project** → **Next**. Name `first_program`, Location `D:\code`, tick **Place solution and project in the same directory** → **Create**.

02 Find Your Way Around

▶ VIDEO 02:45

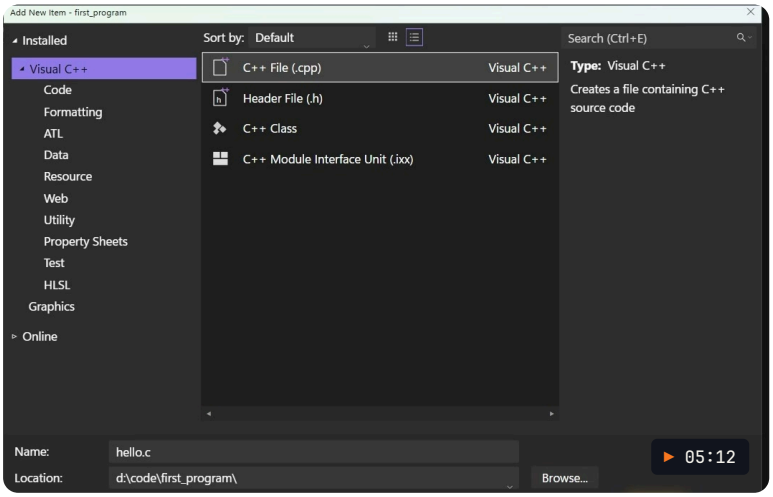
- **Solution Explorer** shows the solution, the project, **Header Files** (`.h`) and **Source Files** (`.c`). Our code goes in Source Files.
- These groups aren't folders on disk. **Show All Files** shows the real folder view.

03 Add hello.c

▶ VIDEO 03:58

2. **Add a source file.** Right-click **Source Files** → **Add** → **New Item** → (**Show All Templates**) → **C++ File**, name it `hello.c` → **Add**.

FIGURE 3.1 **Add New Item: C++ File, named hello.c**



WATCH OUT **The extension decides** `.c` is compiled as C, `.cpp` as C++. The dialog suggests `Source.cpp`: change the name to end in `.c`.

04 Write the Code

▶ VIDEO 05:22

3. **Type the program** exactly, including every bracket, quotation mark and semicolon.

hello.c — C SOURCE

```c
#include <stdio.h>

int main()
{
    printf("hello C\n");
    return 0;
}
```

- **Line 1**: `stdio.h` declares `printf`.
- **Line 3**: the program starts inside `main`.
- **Line 5**: `printf` prints the text; `\n` starts a new line.
- **Line 6**: `return 0;` ends `main` normally.

05 Build and Run

▶ VIDEO 06:30

4. **Build**: **Build** → **Build Solution** (**Ctrl+Shift+B**). Output ends with `Build: 1 succeeded, 0 failed`.
5. **Run**: **Debug** → **Start Without Debugging** (**Ctrl+F5**).

Microsoft Visual Studio Debug Console

```
hello C

...\x64\Debug\first_program.exe exited with code 0.
Press any key to close this window . . .
```

> KEY IDEA **Use Ctrl+F5 for now**
>
> It builds if needed, then runs without the debugger. "exited with code 0" is the `0` from `return 0`.

06 What's on Disk

▶ VIDEO 08:02

6. Right-click the project → **Open Folder in File Explorer**. The program is `x64\Debug\first_program.exe`, **named after the project**, not the source file.

CODE 1 file on GitHub github.com/abitclearer/c-programming-from-scratch/tree/main/03-your-first-c-program